Microsoft Tunnel for MAM — Mobile Access to Intranet Resources — Government (GCC High)

Implementation Solution Guide

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Contents

Executive Summary — Microsoft Tunnel for MAM (Mobile Access to Intranet Resources) — Government (GCC High) Edition 4

The business problem . . . 4

The solution . . . 4

Availability finding (important) . . . 4

What the customer receives . . . 4

Business outcome . . . 5

Solution Overview — Microsoft Tunnel for MAM (Mobile Access to Intranet Resources) — Government (GCC High) Edition 5

Availability — read this first . . . 5

What this solution delivers . . . 6

Commercial vs GCC High — what's different . . . 7

Who this is for . . . 8

Requirements Guide — Microsoft Tunnel for MAM — Government (GCC High) Edition 8

R1. GCC High or DoD tenant (US sovereign cloud) . . . 8

R2. Intune Plan 2 (Government) or equivalent add-on . . . 9

R3. Intune Administrator role (Government portal) . . . 9

R4. Microsoft Tunnel Gateway — Linux host . . . 9

R5. TLS server certificate (from a non-Cloud-PKI source) . . . 9

R6. Network — inbound/outbound for the gateway . . . 10

R7. Public DNS + publicly routable entry point . . . 10

R8. Required mobile apps (BYOD devices) . . . 10

R9. Trusted-root certificate profile (if on-prem resources use a private CA) . . . 10

R10. Line-of-business (LOB) app MAM SDK integration (only if publishing internal apps) . . . 11

R11. Pilot security group (Government Entra) . . . 11

Global Requirements — Intune Solutions (category-level) 11

G1. Microsoft Entra ID tenant . . . 12

G2. Microsoft Intune subscription . . . 12

G3. Administrative roles . . . 12

G4. Device enrollment in Intune . . . 12

G5. Pilot security group . . . 12

G6. Supported Windows baseline . . . 12

G7. Network access to Microsoft cloud endpoints . . . 12

Implementation Guide — Microsoft Tunnel for MAM — Government (GCC High) Edition 13

Phase 1 — Assess (gate 0) . . . 13

Phase 2 — Configure . . . 13
Phase 3 — Pilot . . . 17
Phase 4 — Validate . . . 18
Phase 5 — Expand . . . 18
Government install & endpoint quick-reference . . . 18
Testing & Validation — Microsoft Tunnel for MAM — Government (GCC High) Edition 19
A. Automated validation (control plane) . . . 19
B. Manual device tests (data plane) . . . 20
C. Government-specific verification . . . 21
Troubleshooting — Microsoft Tunnel for MAM — Government (GCC High) Edition 21
Government-specific issues **[GOV]** . . . 21
Gateway / infrastructure . . . 22
Android device issues . . . 23
iOS/iPadOS device issues . . . 23
General . . . 24
Best Practices — Microsoft Tunnel for MAM — Government (GCC High) Edition 24
Deployment discipline . . . 25
Government correctness **[GOV]** . . . 25
Security & compliance **[GOV]** . . . 25
Certificates & PKI **[GOV]** . . . 26
Networking & performance . . . 26
Platform gotchas . . . 26
Operations . . . 26
Deliverables Summary — Microsoft Tunnel for MAM — Government (GCC High) Edition 27
Documents (source) . . . 27
PowerShell scripts (`scripts/`) . . . 28
Named objects this solution creates . . . 29
Deployment model . . . 29
Appendix A — PowerShell Scripts 30

Executive Summary — Microsoft Tunnel for MAM (Mobile Access to Intranet Resources) — Government (GCC High) Edition

Solution code: TUNMAMG · **Platform:** Microsoft Intune · **Cloud:** US Sovereign (GCC High / DoD)

The business problem

Government agencies, DoD components, and defense-industrial-base contractors need staff to reach **internal, on-premises applications** from **personal mobile devices** — without enrolling those devices in full management and without standing up costly legacy VPN infrastructure. The obstacle in the sovereign cloud is that **Government (GCC High / DoD) is not Commercial with a different sign-in**: the admin portals, service endpoints, and gateway install differ, and a Commercial runbook applied to a Government tenant **silently fails** (most often a Tunnel Gateway that installs but never appears in the admin center).

The solution

This is the **Government (GCC High) edition** of the Microsoft Tunnel for MAM deployment. **Microsoft Tunnel for MAM** extends the Microsoft Tunnel VPN Gateway to **unenrolled Android and iOS/iPadOS devices**, giving users secure, app-scoped access to intranet resources through **Microsoft Edge and MAM-integrated line-of-business apps** — with modern authentication, single sign-on, Conditional Access, and app protection — while IT never takes control of the personal device.

The solution is packaged specifically for the US sovereign cloud: the correct `*.us` portals, the `intune_env=FXP` Government gateway install flag, **US Gov Microsoft Graph** endpoints in every script (`-Environment USGov` / `USGovDoD`), and every **Commercial-vs-Government parity gap** documented at the step where it matters.

Availability finding (important)

Tunnel for MAM is fully supported in both GCC High and DoD — Microsoft lists it among the Intune advanced capabilities available in those environments (verified 2026). This is not a feature you must work around in Government. **However, it does not support FIPS** on the Android or iOS data path — a compliance-relevant constraint that must be reviewed and accepted by your ISSO/ISSM before deployment. Certificate issuance should plan on an **enterprise or public CA**, because **Cloud PKI is GCC-High-only and still "planned."**

What the customer receives

- **Seven implementation documents** — overview (with a Commercial-vs-GCC-High comparison and availability caveat up front), requirements, click-by-click implementation in the Government portals, testing with pass/fail gates, troubleshooting, best practices, and a cited **GCC High parity reference**.
- **Five PowerShell scripts** — Discovery, Validation, Report (read-only), plus safe Remediation and Scope-Targeting (default `-WhatIf`) — all connecting to **US Gov Graph**.
- A gated deployment path — **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** — so the design is proven on a pilot before it scales.

Business outcome

A **correctly configured** Tunnel-for-MAM deployment on a GCC High / DoD tenant: BYOD access to intranet resources without device enrollment, the sovereign endpoints right the first time, parity gaps handled, and the no-FIPS decision made consciously — delivered self-guided at a fraction of the cost of a bespoke consulting engagement.

White-label solution. Verify all GCC-High/DoD specifics against current Microsoft US Government documentation at deployment time.

Solution Overview — Microsoft Tunnel for MAM (Mobile Access to Intranet Resources) — Government (GCC High) Edition

Solution code: TUNMAMG · **Platform:** Microsoft Intune · **Cloud:** US Sovereign (GCC High / DoD) · **Tier:** T4

Availability — read this first

Good news, with caveats. Microsoft Tunnel Gateway **and** Microsoft Tunnel for Mobile Application Management (MAM) **are both supported** in the US sovereign clouds — **US Government Community Cloud (GCC) High** *and* **US Department of Defense (DoD)**. This is not a case where a Commercial networking feature is missing in Government: Microsoft explicitly lists **Microsoft Tunnel for Mobile Application Management** among the Intune advanced capabilities that support GCC High and DoD.

> *Source: *Microsoft Intune Government Service overview* (updated 2026-06-04) — "The following Intune advanced capabilities support the GCC High and DoD environments: … Microsoft Tunnel for Mobile Application Management." ([learn.microsoft.com](https://learn.microsoft.com/en-us/intune/intune-service/fundamentals/intune-govt-service-description)) Source: *Use Microsoft Tunnel VPN with Android/iOS devices that don't enroll with Intune* — "Microsoft Tunnel for MAM on Android/iOS is supported with the following sovereign cloud environments: U.S. Government Community Cloud (GCC) High, U.S. Department of Defense (DoD)." ([Android](https://learn.microsoft.com/en-us/intune/device-security/microsoft-tunnel/mam-android) · [iOS](https://learn.microsoft.com/en-us/intune/device-security/microsoft-tunnel/mam-ios))*

But three Government-specific constraints shape the whole deployment — know them before you buy hardware or write a design:

1. **No FIPS.** *Microsoft Tunnel for MAM on Android and on iOS/iPadOS does **not** support Federal Information Processing Standard (FIPS).* If your ATO/STIG obligations mandate FIPS 140-validated cryptography for this data path, Tunnel for MAM is **not** the compliant control — flag this to your ISSO/ISSM before scoping. (Verified on both the Android and iOS Microsoft Learn pages.)
2. **Not on 21Vianet.** Tunnel for MAM is *not* supported on Azure operated by 21Vianet (China). Irrelevant to US Gov tenants, but recorded for completeness.

3. **Certificate-authority parity gap.** Microsoft **Cloud PKI** (the cloud-hosted CA advanced capability) is **GCC High only and still on the "planned" list** — it is **not yet generally available in DoD**, and even in GCC High you should **verify current status** before depending on it. Because Tunnel for MAM relies on a **TLS server certificate** (and often **trusted-root** profiles for on-prem resources), plan your PKI on an on-premises/enterprise CA or a supported public CA rather than assuming Cloud PKI. (Source: *Government Service overview*, "Intune features planned for GCC High and DoD" table.)

The rest of this solution assumes GCC High as the primary target and calls out DoD differences where they exist.

What this solution delivers

Microsoft Tunnel for MAM lets users on **personal, unenrolled** Android and iOS/iPadOS devices reach your **on-premises intranet apps and resources** (internal web apps, line-of-business services) over a secure VPN — **without** enrolling the device in MDM and **without** giving IT control of the whole device. Access is scoped to **managed apps** (Microsoft Edge and your MAM-integrated line-of-business apps), protected by **modern authentication, single sign-on, Conditional Access, and app protection policies**.

It extends the **Microsoft Tunnel VPN Gateway** — a container that runs on a Linux server you host — to unenrolled devices. You must have the Tunnel Gateway deployed first; Tunnel for MAM is a set of **app configuration + app protection policies** layered on top of that gateway.

This edition packages the deployment for a **GCC High (or DoD) tenant**: the correct `*.us` sovereign admin portals, the `intune_env=FXP` government install flag for the gateway, US Gov Microsoft Graph endpoints (`-Environment USGov` / `-Environment USGovDoD`) in every script, and the Commercial-vs-Government parity gaps documented inline so a government tenant is configured correctly — not with Commercial steps that silently fail.

The deployment model

Every step in this solution follows the standard gated model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with a go/no-go gate between each phase so you never scale a misconfiguration.

What's in the box

- **This Solution Overview** and a **Requirements Guide** (`02-requirements.md`) with verify + implement steps for every prerequisite.
- **Implementation Guide** (`03-implementation.md`) — ordered, click-by-click steps in the GCC High portals with Government caveats inline.
- **Testing & Validation** (`05-testing.md`) with pass/fail gates.
- **Troubleshooting** (`06-troubleshooting.md`) with GCC-High-specific gotchas.
- **Best Practices** (`07-best-practices.md`), an **Executive Summary**, a **GCC High parity reference** (`gcc-high-parity-notes.md`), and a **deliverables summary**.
- **5 PowerShell scripts** — Discovery, Validation, Report, Remediation, and Set-ScopeTargeting — all connecting to **US Gov Microsoft Graph** (`-Environment USGov`).

Commercial vs GCC High — what's different

If you have deployed Microsoft Tunnel for MAM in a Commercial tenant, here is the delta for Government. **The product and the click-path are essentially the same; the URLs, endpoints, and a handful of parity constraints differ.**

Area	Commercial	GCC High / DoD (this edition)
Feature availability	Supported	**Supported** (advanced capability) in **both GCC High and DoD**. Verified 2026-06-04.
Intune admin center	`intune.microsoft.com` / `endpoint.microsoft.com`	`intune.microsoft.us`
Azure portal (Linux VM, networking)	`portal.azure.com`	`portal.azure.us` (Azure Government)
Entra admin center (app registration, groups)	`entra.microsoft.com`	**Entra ID Government** via `portal.azure.us` / Entra Gov endpoints
Sign-in authority	`login.microsoftonline.com`	`login.microsoftonline.us` (both GCC High and DoD)
Microsoft Graph	`graph.microsoft.com` (`-Environment Global`)	**GCC High:** `graph.microsoft.us` (`-Environment USGov`) · **DoD:** `dod-graph.microsoft.us` (`-Environment USGovDoD`)
Tunnel Gateway install	`sudo ./mstunnel-setup`	`sudo intune_env=FXP ./mstunnel-setup` (the `intune_env=FXP` flag points the gateway at the Government cloud)
Device login page	`microsoft.com/devicelogin`	`microsoft.com/devicelogin` (same URL; authenticates against the Gov authority)
FIPS	(Tunnel for MAM does not support FIPS in either cloud)	**No FIPS support** — a compliance-relevant gap in Government.
Cloud PKI (for TLS/trusted-root certs)	Generally available	**GCC High only, "planned"** (verify status); **not in DoD.** Use enterprise/public CA.
Licensing	Intune Plan 2 or add-on	Same requirement; ensure the **Government** SKU (G3/G5 Gov, Intune Plan 2 Gov) carries it.
STIG hardening	Optional	**Expected** — DISA MDM/EMM STIGs apply to the tenant and the Linux host.

The single most common Government failure: running the gateway installer **without** `intune_env=FXP`. The gateway then tries to register against the Commercial Intune service and **never appears** in your `intune.microsoft.us` admin center. See `03-implementation.md` and `06-troubleshooting.md`.

The second most common: admins bookmark or automate against `*.com` portals/Graph out of habit; in Government every plane is `*.us`. Every script in this package defaults to the US Gov endpoints.

Who this is for

Government agencies, DoD components, and defense-industrial-base (DIB / CMMC) contractors on a **GCC High or DoD** Microsoft 365 tenant who need **BYOD / unenrolled** mobile access to internal, on-premises resources without full device enrollment — and who need it configured correctly for the sovereign cloud the first time.

White-label solution. Verify all GCC-High/DoD specifics against current Microsoft US Government documentation at deployment time; Microsoft updates Government feature parity frequently. Items marked "verify for GCC High/DoD" should be confirmed live in your tenant.

Requirements Guide — Microsoft Tunnel for MAM — Government (GCC High) Edition

Solution code: TUNMAMG

Confirm every requirement below before you configure anything. This list is **in addition to** the category-level prerequisites in `global-requirements-intune-solutions.md` (Entra tenant, Intune subscription, admin roles, pilot group, network access) — but note that for Government you verify those in the `*.us` portals, not `*.com`.

Each requirement follows: **Description · Why · Verify · Implement-if-missing.** Where a detail is Government-specific and not fully documented, it is flagged **"verify for GCC High/DoD."**

R1. GCC High or DoD tenant (US sovereign cloud)

Description: An active Microsoft 365 **Government** tenant — GCC High or DoD (built on Azure Government, IL4/IL5). Note: "GCC" (standard) is a *different* offering and rides the commercial Intune instance; this solution targets **GCC High / DoD**. **Why:** The sovereign portals, endpoints, and the `intune_env=FXP` install flag only apply to the Government cloud. Commercial or GCC-standard steps will not match. **Verify:** Sign in at `https://portal.azure.us` and `https://intune.microsoft.us`. Your sign-in should redirect through `login.microsoftonline.us`. In the Microsoft 365 admin center, the tenant's environment shows a Government plan. **Implement:** Government tenants are provisioned through Microsoft's Government onboarding / an authorized Government reseller with eligibility validation — not self-service. Engage your Microsoft Government account team.

R2. Intune Plan 2 (Government) or equivalent add-on

Description: Tunnel for MAM is an **Intune advanced capability**. It requires **Microsoft Intune Plan 2** *or* the **Intune Suite** / an additional subscription — in the **Government** SKU. **Why:** Without the advanced-capability license, the Tunnel-for-MAM policies and the Defender tunnel client return "Invalid Licensing." (Verified: both Android and iOS MAM pages state "This feature requires Microsoft Intune Plan 2 or an additional subscription.") **Verify:** In `intune.microsoft.us` → **Tenant administration** → **Connectors and tokens / Licensing**, confirm the advanced-capability entitlement. Confirm your users are assigned a Government license that carries Intune Plan 2 (e.g., **Microsoft 365 G5 Gov**, or **Intune Plan 1 Gov + Intune Plan 2 Gov add-on** / **Intune Suite Gov**). **Verify exact SKU availability for GCC High/DoD** with your licensing desk — Government SKU names and bundling differ from Commercial. **Implement:** Assign the Government Intune Plan 2 / Suite license to pilot users, then all target users.

R3. Intune Administrator role (Government portal)

Description: The operator account holds the **Intune Administrator** Entra role and an Intune license. The account that **registers the Tunnel Gateway** must be Intune Administrator (or equivalent) *and* licensed. **Why:** Creating server/site configs, app configuration/protection policies, and completing gateway registration all require elevated Intune rights. **Verify:** In Government Entra (via `portal.azure.us` → Microsoft Entra ID → **Roles and administrators**), confirm **Intune Administrator**. Confirm the same account has an Intune license. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes only.

R4. Microsoft Tunnel Gateway — Linux host

Description: A **Linux server** (on-premises or in Azure Government) that runs a container hosting the Tunnel Gateway software. **Why:** Tunnel for MAM extends an existing Tunnel Gateway; the gateway is the VPN termination point that bridges devices to your intranet. **Verify:** Supported OS/container (per Microsoft prerequisites): **Red Hat Enterprise Linux (RHEL) 8.9–9.7 / 10.0 with Podman**, or **Ubuntu 22.04 / 24.04 with Docker CE**; **64-bit AMD/Intel**. Sizing example: **~1,000 devices = 4 CPU, 4 GB RAM, 30 GB disk, 1 server** (scale out with more servers behind a load balancer). Confirm the OS is a supported, patched build. **Implement:** Stand up the Linux host, install Podman (RHEL) or Docker CE (Ubuntu), and apply your STIG hardening baseline *before* installing Tunnel. (Install steps are in `03-implementation.md`.)

> ***Verify for GCC High/DoD:*** *exact supported RHEL/Ubuntu/Podman/Docker versions move over time — confirm against the current *Microsoft Tunnel prerequisites* page at deployment. ([learn.microsoft.com](https://learn.microsoft.com/en-us/intune/device-security/microsoft-tunnel/prerequisites))*

R5. TLS server certificate (from a non-Cloud-PKI source)

Description: A trusted **TLS certificate** (PEM `site.crt+site.key`, or `site.pfx`) whose **Subject Alternative Name (SAN)** matches the gateway's **public FQDN or IP**. Key **2048-bit or larger**. OCSP/CRL endpoint reachable from the server. **Why:** The TLS cert secures the device-to-gateway tunnel. iOS caps public certs at **398 days** (admin-installed roots up to 730); Android recommends 398 days max. **Verify:** Certificate SAN = gateway FQDN/IP; chain is complete (root + intermediates); not expired; revocation endpoint reachable. **Implement:** Issue from your **enterprise/on-premises CA** or a **supported public CA**.

> ***Government caveat:*** *Do **not** assume **Microsoft Cloud PKI** — it is **GCC High only and "planned"** (not GA in DoD; verify status in GCC High). Plan certificate issuance on an enterprise or public CA. If your on-prem resources use a **private CA**, you will also need **trusted-root certificate profiles** (see R9).*

R6. Network — inbound/outbound for the gateway

Description: Firewall openings for the Tunnel Gateway. **Why:** Devices reach the gateway inbound; the gateway reaches Intune/Entra/Graph outbound. **Verify / Implement:**

- **Inbound to gateway: TCP 443** and **UDP 443** (UDP improves performance via DTLS; you may disable UDP and force TCP). Optional **TCP 22** for SSH admin.
- **Outbound from gateway (TCP 443):** the Microsoft Tunnel service FQDNs — `*.manage.microsoft.com`, `login.microsoftonline.com`, `graph.microsoft.com`, `*.sts.windows.net`, `*.blob.core.windows.net` / `*.blob.storage.azure.net`, `powerlift-frontdesk.acompli.net`.

> ***Verify for GCC High/DoD:*** *in the sovereign cloud several of these resolve to* `*.us` *Government endpoints (e.g., Government Graph/login). Allow the* ***US Government*** *Office 365 / Intune URL & IP ranges published for GCC High/DoD, not only the Commercial list. Confirm the current Government endpoint list before locking firewall rules.*

R7. Public DNS + publicly routable entry point

Description: The gateway **Site** uses a **public IP or FQDN** that resolves in **public DNS** to a **publicly routable** address (directly or via a load balancer). **Why:** Unenrolled BYOD devices connect from the internet; the site endpoint must be reachable and match the TLS SAN. **Verify:** `nslookup <gateway-fqdn>` from the public internet returns the expected address. **Implement:** Create the public DNS record; ensure the TLS cert SAN matches it exactly.

R8. Required mobile apps (BYOD devices)

Description: The apps users must have on their **unenrolled** device. **Why:** These deliver the tunnel client and the managed browser. **Verify / Implement:**

- **Android Enterprise: Microsoft Defender** (the tunnel client — must be **v1.0.4722.0101+**), **Microsoft Edge**, and **Company Portal** (installed but **no sign-in / no enrollment** required). Install manually from Google Play; on unenrolled devices these often will **not** surface in Company Portal.
- **iOS/iPadOS: No Company Portal or Defender requirement.** The VPN is provided by the **Tunnel for MAM iOS SDK** compiled into each managed app (and Microsoft Edge). Users need **Microsoft Edge** and your MAM-integrated LOB apps.

> ***Verify for GCC High/DoD:*** *confirm the Government-cloud builds of Defender/Edge/Company Portal support Tunnel for MAM in your tenant.*

R9. Trusted-root certificate profile (if on-prem resources use a private CA)

Description: An Intune **trusted certificate profile** carrying your private CA **root and all intermediates**. **Why:** Edge/LOB apps must trust the on-prem server's TLS chain. Android requires **MAM SDK 9.5.0+** for trusted-root support; iOS SDK requires **DER-encoded X.509 or PEM**. Android can fail chain-building if intermediates are missing — deploy the **full chain**. **Verify:** Profile contains root + every intermediate; format is DER/PEM. **Implement:** Create the trusted certificate profile in `intune.microsoft.us` and reference it from the app configuration policy (steps in `03-implementation.md`).

R10. Line-of-business (LOB) app MAM SDK integration (only if publishing internal apps)

Description: Internal iOS/Android apps that will use the tunnel must integrate the **Intune App SDK + MSAL**, and — for iOS — the **Tunnel for MAM iOS SDK**; for Android, MAM SDK **9.5.0+** for trusted roots. **Why:** The tunnel is scoped to MAM-managed apps; an un-integrated app cannot use it. **Verify:** App builds link the required SDK versions (iOS: MAM-SDK 16.1.1+, MSAL 1.2.3+, Xcode 14+); Entra app registration matches the app's Bundle/Package ID. **Implement:** Hand the SDK integration to your app developers (out of scope for infrastructure admins). Microsoft Edge already supports the tunnel with no code work — many agencies pilot with **Edge-only** first.

R11. Pilot security group (Government Entra)

Description: An Entra **security group** of pilot BYOD users (per the category global requirement, in the Government tenant). **Why:** All three MAM policies must target the **same** groups; you pilot before expanding. **Verify:** Group `GRP-TUNMAMG-Pilot` exists in Government Entra. **Implement:** Create it (or let the included `TUNMAMG-Set-ScopeTargeting.ps1` build/populate it against US Gov Graph). Naming and reuse are covered in `03-implementation.md`.

Naming convention used throughout this solution

Reuse these names so every artifact is traceable:

Artifact	Name
Server configuration	`SC-TUNMAMG-Gov`
Tunnel Site	`SITE-TUNMAMG-Gov`
App config policy — Defender (Android)	`AC-TUNMAMG-Defender-Android`
App config policy — Edge (Android)	`AC-TUNMAMG-Edge-Android`
App protection policy — Edge (Android)	`AP-TUNMAMG-Edge-Android`
App config policy — Edge/LOB (iOS)	`AC-TUNMAMG-Edge-iOS`
App protection policy — iOS	`AP-TUNMAMG-iOS`
Trusted root profile	`CERT-TUNMAMG-Root`
Pilot group	`GRP-TUNMAMG-Pilot`
Production group	`GRP-TUNMAMG-Prod`

Verify all GCC-High/DoD specifics against current Microsoft US Government documentation at deployment time.

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution.

Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

Implementation Guide — Microsoft Tunnel for MAM — Government (GCC High) Edition

Solution code: TUNMAMG · **Model:** Assess → Configure → Pilot → Validate → Expand

This guide is written for a **novice administrator**. It gives full navigation in the **Government** (`*.us`) portals and calls out **GCC-High/DoD caveats inline at the step where they matter**. Reuse the object names from `02-requirements.md`.

> ***Portals you will use (Government):*** *-* ***Intune admin center:*** `https://intune.microsoft.us` *-* ***Azure Government portal:*** `https://portal.azure.us` *(Linux VM, networking, Entra ID Government) -* ***Sign-in authority:*** `https://login.microsoftonline.us` *-* ***Device-login (gateway registration):*** `https://microsoft.com/devicelogin`

Phase 1 — Assess (gate 0)

1. Work through `02-requirements.md`. Do not proceed until R1–R8 pass.
2. **Run the Discovery script** from this package to snapshot the Government tenant:

```
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Discovery.ps1
```

It connects to **US Gov Graph** (`-Environment USGov`) and reports existing Tunnel sites/server configs, Tunnel-for-MAM app config/protection policies, and target groups.

1. **Compliance pre-check (critical):** confirm with your **ISSO/ISSM** that **no-FIPS** on the Tunnel-for-MAM data path is acceptable for the data classification you intend to carry. *Tunnel for MAM does not support FIPS on Android or iOS.* If FIPS is mandated, **stop** — this is not the compliant control.

Gate 0 → **proceed only if:** requirements pass, licensing (Intune Plan 2 Gov) confirmed, and the no-FIPS decision is signed off.

Phase 2 — Configure

You will (A) install the Tunnel Gateway with the **Government install flag**, then (B) create the Server configuration and Site, then (C) layer the Tunnel-for-MAM policies.

A. Install the Microsoft Tunnel Gateway (Government)

> ■■ ***The #1 Government mistake:*** *installing without* `intune_env=FXP`*. The gateway then registers against* ***Commercial*** *Intune and never appears in* `intune.microsoft.us`*. Do not skip the flag.*

1. On the prepared Linux host (R4), download the installer:

```
wget --output-document=mstunnel-setup https://aka.ms/microsofttunneldownload
```

1. **Run the installer with the Government cloud flag:**

```
sudo intune_env=FXP ./mstunnel-setup
```

The `intune_env=FXP` variable points the gateway at the **US Government** cloud. (Microsoft's documented Government sequence: run `sudo ./mstunnel-setup`, then `sudo intune_env=FXP`

`./mstunnel-setup`.) For a rootless Podman container, prefix `mst_rootless_mode=1` as well.

1. Accept the **EULA** when prompted.

2. When prompted, install the **TLS certificate** (R5):

• **PEM:** full chain → `/etc/mstunnel/certs/site.crt`; private key → `/etc/mstunnel/private/site.key`.

• **PFX:** → `/etc/mstunnel/private/site.pfx`.

1. **Register with Intune:** the script shows a device code. Browse to `https://microsoft.com/devicelogin`, enter the code, and sign in with an **Intune Administrator** account that has an Intune license. Because your tenant is Government, this authenticates against `login.microsoftonline.us` automatically.

2. When prompted for a **Site GUID**, you will pick the Site you create in step B (you can create the Server config + Site first, then return — the script resumes where it left off).

3. On RHEL, finish with `mst-cli server restart`.

Caveat — proxy/firewall: *if egress passes an SSL-inspecting proxy, add trusted roots to* `/etc/mstunnel/ca-trust` *and configure the proxy per Microsoft's install doc. In Government, ensure your firewall allows the* ***US Gov*** *Intune/Entra/Graph endpoints (R6), not only the Commercial ranges.*

B. Create the Server configuration and Site (in `intune.microsoft.us`*)*

Server configuration → name it `SC-TUNMAMG-Gov`**:**

1. Sign in to `https://intune.microsoft.us`.

2. Go to **Tenant administration → Microsoft Tunnel Gateway → Server configurations tab → Create new**.

3. **Basics:** Name = `SC-TUNMAMG-Gov`. **Next**.

4. **Settings:**

• **IP address range:** a range that does **not** conflict with your intranet (Microsoft suggests the APIPA range `169.254.0.0/16`).

• **Server port:** e.g. `443`.

• **DNS servers:** your internal DNS (so tunnel clients resolve intranet names). Add the DNS server IPs to **split-tunnel include** rules if you use corporate DNS.

• **DNS suffix search** (optional): your internal domain.

• **Split tunneling:** add **include** rules for the intranet subnets/hosts BYOD users must reach (e.g., `192.168.0.0/16`). **Never** use a `0.0.0.0` include/exclude — the gateway cannot route with it.

1. **Review + create → Create.**

Site → name it `SITE-TUNMAMG-Gov`**:**

1. **Tenant administration → Microsoft Tunnel Gateway → Sites tab → Create.**

2. **Name** = `SITE-TUNMAMG-Gov`.

3. **Public IP address or FQDN:** the public entry point (R7) — must be resolvable in public DNS and match the TLS SAN.

4. **Server configuration:** select `SC-TUNMAMG-Gov`.

5. **URL for internal network access check:** an internal HTTP(S) URL the server can reach every 5 minutes to prove intranet connectivity.

6. Set upgrade/maintenance-window options as your change process requires. **Create.**

7. Return to the Linux installer (step A.6) and enter this Site's GUID. Confirm the server shows **online** under **Microsoft Tunnel Gateway** → **Health status**.

Gate — gateway: server is **online** and **Internal network accessibility** is healthy before you touch MAM policies.

C. Configure Tunnel-for-MAM policies

You deploy **three** policies **per platform**, all targeting the **same** group (`GRP-TUNMAMG-Pilot` first). For Android that's two app config policies (Defender, Edge) + one app protection policy (Edge). For iOS that's an app config policy (Edge/LOB) + app protection policy, plus optional trusted-cert profile.

> ***Caveat — enrolled devices:*** *a device may use **either** MDM Tunnel **or** MAM Tunnel, **never both**. Do not deploy MAM Tunnel app-config settings to a device that already has an MDM Tunnel VPN profile — it causes client networking failures.*

C-Android-1 · App configuration policy — Defender (tunnel client)

Name: `AC-TUNMAMG-Defender-Android`

1. In `intune.microsoft.us` → **Apps** → **Configuration** → **Create** → **Managed Apps**.

2. **Basics:** Name = `AC-TUNMAMG-Defender-Android`. **Select public apps** → **Microsoft Defender Endpoint** (Android) → **Select** → **Next**.

3. **Settings** → **Microsoft Tunnel settings:**

• **Use Microsoft Tunnel VPN** = **Yes**.

• **Connection name:** e.g. `mam-tunnel-vpn`.

• **Select a site** → `SITE-TUNMAMG-Gov`.

• **Per-App VPN (Android)** *(optional but recommended)*: add **Microsoft Edge** (include Edge so identity-switch and notifications work). Add other managed apps as needed.

• Do **not** enable **Always-on VPN** — MAM Tunnel for Android does **not** support it; use **Strict Tunnel Mode** via the Edge policy instead to block traffic when the tunnel is down.

1. **Assignments** → add `GRP-TUNMAMG-Pilot`. **Next** → **Create.**

> ■■ *Target only **one** Defender app config policy per device, pointing at a **single** Site. Multiple Defender policies with different sites cause a race condition and tunnel failure.*

C-Android-2 · App configuration policy — Edge (identity switch / strict tunnel)

Name: `AC-TUNMAMG-Edge-Android`

1. **Apps** → **Configuration** → **Create** → **Managed Apps.** Name = `AC-TUNMAMG-Edge-Android`. Public app = **Microsoft Edge** (Android). **Next.**

2. **Settings** → **General configuration settings** — add these **Name/Value** pairs (no trailing spaces):

Name	Value	Effect
`com.microsoft.intune.mam.managedbrowser.StrictTunnelMode`	`True`	Blocks Edge internet traffic when the MAM tunnel is down
`com.microsoft.intune.mam.managedbrowser.TunnelAvailable.IntuneMAMOnly`	`True`	Identity switch: connect VPN on work account, disconnect on personal/in-private

1. **Assignments** → `GRP-TUNMAMG-Pilot` (same group as the Defender policy). **Create.**

C-Android-3 · App protection policy — Edge (auto-start tunnel)

Name: `AP-TUNMAMG-Edge-Android`

1. **Apps** → **Protection** → **Create** → **Android.** Name = `AP-TUNMAMG-Edge-Android`. **Next.**
2. **Apps:** Select public apps → **Microsoft Edge** → **Next.**
3. **Data protection:** scroll to bottom → **Start Microsoft Tunnel connection on app-launch** = **Yes**. **Next.**
4. Pass through **Access requirements** and **Conditional launch** (set to your data-protection standard).
5. **Assignments** → `GRP-TUNMAMG-Pilot` (same group). **Create.**

When all three Android policies target the same group, launching Edge automatically connects the tunnel.

C-iOS-1 · App configuration policy — Edge / LOB

Name: `AC-TUNMAMG-Edge-iOS`

1. **Apps** → **Configuration** → **Create** → **Managed Apps.** Name = `AC-TUNMAMG-Edge-iOS`.

• For **Edge**: **Select public apps** → **Microsoft Edge for iOS/iPadOS.**

• For **LOB apps**: **+ Select custom apps** → enter the app's **Bundle ID**, Platform **iOS/iPadOS** → **Add**.

1. **Settings** → **Microsoft Tunnel for Mobile Application Management settings:**

• **Use Microsoft Tunnel for MAM** = **Yes**.

• **Connection name:** `mam-tunnel-vpn`.

• **Select a Site** → `SITE-TUNMAMG-Gov`.

• **Root Certificate** (optional): select `CERT-TUNMAMG-Root` if reaching private-CA-protected resources.

1. **Edge identity switch / strict tunnel** — add the same two Name/Value pairs as Android (`StrictTunnelMode` = `True`, `TunnelAvailable.IntuneMAMOnly` = `True`).

2. **Federated Entra tenants only** — add General config Name `com.microsoft.tunnel.custom_configuration`, Value `{"bypassedUrls":["<your-federation-STS-host>"]}` so sign-in bypasses the tunnel for the STS. *(Many Government tenants are federated — do not skip this if yours is.)*

3. **Assignments** → `GRP-TUNMAMG-Pilot`. **Create.**

C-iOS-2 · App protection policy — iOS

Name: `AP-TUNMAMG-iOS`

1. **Apps** → **Protection** → **+ Create policy** → **iOS/iPadOS.** Name = `AP-TUNMAMG-iOS`. **Next.**

2. **Apps:** add **Microsoft Edge** (public) and/or your LOB app by **Bundle ID** (+ Select custom apps).

3. Configure **Data protection / Access requirements / Conditional launch** to your standard.

4. **Assignments** → `GRP-TUNMAMG-Pilot` (same group). **Create.**

> ***iOS reality check:*** *on iOS the tunnel is **Per-App only** (no device-wide VPN), delivered by the **Tunnel for MAM iOS SDK** — so **LOB apps must be recompiled** with the SDK (Edge works out of the box). VPN connections **do not appear** in iOS Settings. Edge on iOS does **not** support Kerberos or certificate-based auth to on-prem sites (NTLM SSO is supported for Edge, not LOB).*

C-optional · Trusted-root certificate profile

Name: `CERT-TUNMAMG-Root` — needed only if on-prem resources use a **private CA**.

1. **Devices** → **Configuration** → **Create** → **New Policy** → platform (any; a trusted-cert profile for any platform is accepted by Tunnel for MAM) → **Templates** → **Trusted certificate.**

2. Upload the **root and all intermediate** certs (DER/PEM). Name = `CERT-TUNMAMG-Root`. Create.

3. Reference it from the app config policy's **Root Certificate** selector (C-iOS-1 step 2 / Android app-config Tunnel-for-MAM settings).

> *Deploy the **full chain** (root + intermediates). Android/iOS can fail server-trust if intermediates are missing.*

Phase 3 — Pilot

1. Ensure all policies for the platform you're piloting target `GRP-TUNMAMG-Pilot` only. Use `TUNMAMG-Set-ScopeTargeting.ps1` to build/populate the group (defaults to `-WhatIf`; add `-Commit` to apply).

2. Pilot users install the required apps (R8) manually from the store:

- **Android:** Microsoft Defender (v1.0.4722.0101+), Microsoft Edge, Company Portal (no sign-in).
- **iOS:** Microsoft Edge and any MAM-integrated LOB app.

1. Launch **Microsoft Edge**, sign in with the **work/Government** account, browse to an **internal** URL.

Gate — pilot: run `TUNMAMG-Validation.ps1` (below) — all checks pass — and pilot users reach intranet resources through Edge.

Phase 4 — Validate

Run the validation script and the manual tests in `05-testing.md`:

```
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Validation.ps1
```

It confirms (read-only, US Gov Graph): gateway site/server present and healthy; the three MAM policies exist and target the same group; licensing/advanced-capability present; no MDM-Tunnel + MAM-Tunnel collision on the same group.

Gate — validate: validation clean + intranet access proven + Strict Tunnel Mode blocks traffic when disconnected.

Phase 5 — Expand

1. Create/confirm `GRP-TUNMAMG-Prod` and add production BYOD users (use `TUNMAMG-Set-ScopeTargeting.ps1`).
2. Add `GRP-TUNMAMG-Prod` to the **Assignments** of **all** the platform's policies (keep the group set identical across the three policies).
3. Scale the gateway: add Tunnel servers to `SITE-TUNMAMG-Gov` behind a load balancer as device count grows (each server joins the same Site and inherits `SC-TUNMAMG-Gov`).
4. Generate a point-in-time report:

```
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Report.ps1
```

1. Establish operations: monitor **Health status**, keep servers within one version of current (auto-upgrade or maintenance window), and rotate the TLS cert before expiry (`mst-cli import_cert` → `mst-cli server restart`).

Gate — expand: production group live, reporting captured, cert-rotation and upgrade runbook documented.

Government install & endpoint quick-reference

Item	Value (GCC High)	Value (DoD)
Intune portal	`intune.microsoft.us`	`intune.microsoft.us`
Azure portal	`portal.azure.us`	`portal.azure.us`
Sign-in authority	`login.microsoftonline.us`	`login.microsoftonline.us`
Gateway install flag	`intune_env=FXP`	`intune_env=FXP`
Graph endpoint	`graph.microsoft.us` (`-Environment USGov`)	`dod-graph.microsoft.us` (`-Environment USGovDoD`)

Verify all GCC-High/DoD specifics against current Microsoft US Government documentation at deployment time. Steps marked "verify" should be confirmed live in your tenant.

Testing & Validation — Microsoft Tunnel for MAM — Government (GCC High) Edition

Solution code: TUNMAMG

Run these after Phase 2 (Configure) and before Phase 5 (Expand). Each test has an explicit **Pass / Fail** gate. Combine the automated `TUNMAMG-Validation.ps1` run (US Gov Graph, read-only) with the manual device tests below.

A. Automated validation (control plane)

```
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Validation.ps1
```

Connects with `-Environment USGov` and checks the tenant-side configuration.

#	Check	Pass	Fail action
A1	Graph context is a **US Gov** endpoint (`graph.microsoft.us` / `dod-graph.microsoft.us`)	Context shows `.us` host	You're on Commercial Graph — reconnect with `-Environment USGov/USGovDoD`
A2	Tunnel **Site** `SITE-TUNMAMG-Gov` exists and references `SC-TUNMAMG-Gov`	Both present, linked	Re-create per `03-implementation.md` §B
A3	At least one Tunnel **server** is registered and **online**	Health = online	If missing, the gateway likely installed **without** `intune_env=FXP` — reinstall with the flag
A4	Android: `AC-TUNMAMG-Defender-Android`, `AC-TUNMAMG-Edge-Android`, `AP-TUNMAMG-Edge-Android` exist	All three present	Create the missing policy
A5	iOS: `AC-TUNMAMG-Edge-iOS`, `AP-TUNMAMG-iOS` exist	Present	Create the missing policy
A6	All policies for a platform target the **same** group(s)	Group sets match	Align assignments (use `TUNMAMG-Set-ScopeTargeting.ps1`)
A7	**Single** Defender app-config policy per targeted device, one Site	Exactly one	Remove duplicates (race-condition risk)
A8	No device/group has **both** MDM-Tunnel and MAM-Tunnel settings	No overlap	Separate the groups — collision breaks networking
A9	Intune **Plan 2 / advanced capability** licensing present	Entitled	Assign Government Intune Plan 2 / Suite

Gate A → pass when A1–A9 all pass.

B. Manual device tests (data plane)

Perform on a real pilot BYOD device in `GRP-TUNMAMG-Pilot`.

B1 — Android: apps present & tunnel client valid

• Install **Microsoft Defender (v1.0.4722.0101+)**, **Microsoft Edge**, **Company Portal** from Google Play.

• **Pass:** apps installed; Defender does **not** report "Invalid Licensing." **Fail:** "Invalid Licensing" → install the supported Defender version.

B2 — Android: intranet reachable via Edge

• Open **Edge**, sign in with the **Government work account**, browse to an **internal-only** URL.

• **Pass:** the internal page loads (a brief first-attempt failure right after launch is expected — refresh once). **Fail:** page never loads → check split-tunnel include rules, DNS servers in `SC-TUNMAMG-Gov`, gateway health.

B3 — Android: Strict Tunnel Mode

• Disconnect the tunnel (or take the gateway unreachable). Try to browse in Edge with the work account.

• **Pass:** internet traffic in Edge is **blocked** until the tunnel reconnects. **Fail:** traffic flows → verify `StrictTunnelMode=True` (no trailing space) in `AC-TUNMAMG-Edge-Android`.

B4 — Android: identity switch

• Switch Edge to a **personal** account or in-private.

• **Pass:** VPN disconnects for personal, reconnects for work. **Fail:** verify `TunnelAvailable.IntuneMAMOnly=True`.

B5 — iOS: Edge intranet access (Per-App)

• Install **Edge**, sign in with the work account, browse to an internal URL.

• **Pass:** internal page loads via the SDK-provided Per-App VPN (note: **no VPN entry appears in iOS Settings** — that is expected). **Fail:** check Site selection in `AC-TUNMAMG-Edge-iOS`, and for **federated** tenants confirm the `bypassedUrls` STS entry.

B6 — iOS: LOB app (if applicable)

• Launch an SDK-integrated LOB app; access an on-prem resource.

• **Pass:** resource loads through the tunnel. **Fail:** confirm the app links **Tunnel for MAM iOS SDK + MSAL**, Bundle ID matches the Entra app registration, and the `Tunnel_Allow` API permission has admin consent.

B7 — Private-CA resource trust (if applicable)

• Access an on-prem site secured by your **private CA**.

• **Pass:** no certificate error. **Fail:** deploy `CERT-TUNMAMG-Root` with **root + all intermediates**; on Android, manually install the root on the device if a WebView LOB app still fails.

Gate B → pass when every applicable B-test passes on at least two pilot devices per platform.

C. Government-specific verification

#	Check	Pass
C1	All admin work performed in `*.us` portals (`intune.microsoft.us`, `portal.azure.us`)	Yes
C2	Gateway installed with `intune_env=FXP` and visible in `intune.microsoft.us`	Yes
C3	Scripts connected via **US Gov Graph** (`-Environment USGov` / `USGovDoD`) — no `*.com` calls	Yes
C4	**No-FIPS** limitation reviewed and accepted by ISSO/ISSM for the data carried	Signed off
C5	Firewall allows the **US Government** Intune/Entra/Graph endpoint ranges (not only Commercial)	Yes
C6	Certificate source is enterprise/public CA (**not** assumed Cloud PKI, which is GCC-High-only/planned)	Yes

Gate C → pass when C1–C6 confirmed. Only then proceed to **Expand**.

Verify all GCC-High/DoD specifics against current Microsoft US Government documentation at deployment time.

Troubleshooting — Microsoft Tunnel for MAM — Government (GCC High) Edition

Solution code: TUNMAMG

Symptom → likely cause → fix. Government-specific issues are marked **[GOV]** and appear first because they are the most common cause of "it worked in Commercial but not here."

Government-specific issues [GOV]

The Tunnel Gateway installed successfully but never appears in `intune.microsoft.us`

Cause: The gateway was installed **without** the Government cloud flag, so it registered against the **Commercial** Intune service. **Fix:** Reinstall referencing the Government cloud: `sudo intune_env=FXP ./mstunnel-setup`. Confirm under **Tenant administration → Microsoft Tunnel Gateway → Health**

status in `intune.microsoft.us`. This is the single most frequent Government failure.

Scripts return no data / authenticate to the wrong tenant

Cause: Connected to **Commercial** Graph (`graph.microsoft.com`) instead of US Gov. **Fix:** Reconnect with `Connect-MgGraph -Environment USGov` (GCC High) or `-Environment USGovDoD` (DoD). The provided scripts default to `USGov`; pass DoD parameters for DoD. Verify with `(Get-MgContext).Environment` and confirm the endpoint host ends in `.us`.

Admin can't find Tunnel settings / "the portal looks different"

Cause: Signed in to a `*.com` portal. **Fix:** Use `https://intune.microsoft.us` and `https://portal.azure.us`. Sign-in should route through `login.microsoftonline.us`.

Gateway can reach the internet but can't register with Intune

Cause: Firewall allows only Commercial endpoint ranges. **Fix:** Allow the **US Government** Office 365/Intune/Entra/Graph endpoint ranges (`*.manage.microsoft.com` and the Gov `login`/`graph` `*.us` hosts). Re-verify against the current Government endpoint list.

First-time sign-in fails in Edge/LOB app on a federated Government tenant

Cause: The federation **STS** URL is being routed through the tunnel before authentication completes. **Fix:** In the Edge/LOB app configuration policy add General config key `com.microsoft.tunnel.custom_configuration` = `{"bypassedUrls":["<your-federation-STS-host>"]}`. Many Government tenants are federated — do not skip this.

Compliance flags the deployment for FIPS

Cause: Tunnel for MAM does **not** support FIPS (Android or iOS) — expected behavior, not a misconfiguration. **Fix:** This cannot be enabled. Obtain ISSO/ISSM acceptance for the data classification, or select a FIPS-validated alternative for this path.

Planned to issue certificates from Cloud PKI — it's not available

Cause: Cloud PKI is **GCC High only and "planned"** (not GA in DoD; verify in GCC High). **Fix:** Issue the TLS server cert (and trusted-root material) from an **enterprise/on-premises CA** or a supported **public CA**.

Gateway / infrastructure

Server shows offline or "Internal network accessibility" unhealthy

Cause: Gateway service down, network path blocked, or the internal check URL unreachable. **Fix:** On the host run `mst-cli server restart` (required after install on RHEL 8.4+). Confirm inbound **TCP/UDP 443** reach the server and the site's **internal network access check URL** resolves from the server.

TLS certificate errors when devices connect

Cause: SAN mismatch, incomplete chain, expired cert, or unreachable OCSP/CRL. **Fix:** Ensure the cert **SAN = the site's public FQDN/IP**, the **full chain** is installed (`site.crt` root+intermediates, `site.key` unencrypted), it isn't expired (iOS ≤ 398 days for public certs), and the revocation endpoint is reachable from the server. Update with `mst-cli import_cert` → `mst-cli server restart`.

Upgrades not applying

Cause: Manual-approval upgrade mode or a maintenance window. **Fix:** Approve pending upgrades, or check the site's maintenance-window settings. Stay within one version of the current release.

Android device issues

Defender: "Invalid Licensing, please contact administrator"

Cause: Defender for Endpoint version predates Tunnel support. **Fix:** Install Defender **v1.0.4722.0101+** from Google Play. (Also confirm Intune Plan 2 / advanced-capability licensing is assigned.)

Tunnel doesn't auto-connect when Edge launches

Cause: The three Android policies don't all target the **same** group, or the app protection "Start Microsoft Tunnel connection on app-launch" is not set. **Fix:** Confirm `AC-TUNMAMG-Defender-Android`, `AC-TUNMAMG-Edge-Android`, and `AP-TUNMAMG-Edge-Android` all target the same group; set app-launch start = **Yes** in the protection policy.

Tunnel connects then drops / behaves erratically

Cause: Multiple Defender app-config policies with different Sites (race condition), or Always-on VPN enabled. **Fix:** Target **one** Defender policy per device with **one** Site. Do **not** enable Always-on VPN with MAM Tunnel — use **Strict Tunnel Mode** in the Edge policy instead.

Strict Tunnel Mode not blocking traffic

Cause: `StrictTunnelMode` key missing/mistyped or has a **trailing space**; or device isn't AM-API enrolled (for device-wide strict mode). **Fix:** Set `com.microsoft.intune.mam.managedbrowser.StrictTunnelMode` = `True` with no trailing space in `AC-TUNMAMG-Edge-Android`.

The three required apps aren't visible in Company Portal

Cause: Expected on unenrolled devices. **Fix:** Have users install Defender, Edge, and Company Portal **manually** from Google Play (links in `02-requirements.md`).

Edge can't reach internal resource immediately after launch

Cause: Edge attempts the connection before the tunnel establishes. **Fix:** Refresh the page once the tunnel connects — expected transient behavior.

Private-CA internal site fails (WebView LOB app / Defender cert error)

Cause: Missing intermediates or Defender not yet trusting the private root. **Fix:** Deploy `CERT-TUNMAMG-Root` with **root + all intermediates**; if a WebView LOB app still fails, manually install the root on the device. For Defender's private-CA TLS error, manually install the corresponding root on the device.

iOS/iPadOS device issues

No VPN entry appears in iOS Settings

Cause: Expected — the Tunnel for MAM iOS SDK provides a Per-App VPN scoped to the app's networking layer; it is not a system VPN. **Fix:** None needed. Verify via `mst-cli` active client connections instead.

LOB app can't use the tunnel

Cause: App not integrated with the required SDKs, or Entra app registration mismatch. **Fix:** Ensure the app links **Tunnel for MAM iOS SDK + Intune App SDK + MSAL** (Xcode 14+, MAM-SDK 16.1.1+, MSAL 1.2.3+); the app's **Bundle ID** must match the Entra app registration; grant admin consent to **Microsoft Tunnel Gateway (**`Tunnel_Allow`**)** and **Microsoft Mobile Application Management** API permissions; add the `acct` optional access-token claim.

MFA Conditional Access breaks the iOS tunnel

Cause: Known issue — the Authenticator app doesn't complete the token flow for Tunnel-for-MAM CA MFA. **Fix:** Implement `onTokenRequiredWithCallback` in the LOB app's Tunnel Delegate class.

Edge on iOS can't authenticate to an on-prem site

Cause: Kerberos and certificate-based auth to on-prem sites are **not supported** by Tunnel for MAM on iOS Edge. **Fix:** Use a supported auth method (NTLM SSO is supported for Edge, not LOB apps). No workaround for Kerberos/cert-based.

Both Tunnel VPN and Defender VPN drop on sleep (iOS combined mode)

Cause: "Disconnect on Sleep" without properly configured On-Demand rules. **Fix:** Configure On-Demand rules (Per-App VPN recommended); note SSO conflicts with an all-domains connect rule + Disconnect-on-Sleep.

General

A device using both MDM Tunnel and MAM Tunnel has networking failures

Cause: A device may use **either** MDM Tunnel or MAM Tunnel, never both simultaneously. **Fix:** Separate the groups so no device receives both an MDM Tunnel VPN profile and MAM Tunnel app-config settings.

Newly added custom (LOB) app doesn't show in the policy UI

Cause: Known admin-center UX refresh issue. **Fix:** Refresh the Intune admin center and reopen the policy; the Bundle ID should then appear.

If a symptom isn't listed, re-run `TUNMAMG-Validation.ps1` (US Gov Graph) to localize the fault to control plane vs data plane, then consult the matching section above. Verify all GCC-High/DoD specifics against current Microsoft US Government documentation.

Best Practices — Microsoft Tunnel for MAM — Government (GCC High) Edition

Solution code: TUNMAMG

Field-tested guidance for running Tunnel for MAM correctly on a **GCC High / DoD** tenant. Grouped by theme; Government-specific items are marked **[GOV]**.

Deployment discipline

• **Follow the gates.** Assess → Configure → Pilot → Validate → Expand. Never add production users until pilot validation (`05-testing.md`) is clean.

• **Edge-first pilot.** Microsoft Edge uses the tunnel with **no code changes**. Pilot with Edge to an internal web app before investing in LOB SDK integration — it de-risks the infrastructure independently of app development.

• **Reuse the naming convention** (`SC-`, `SITE-`, `AC-`, `AP-`, `CERT-`, `GRP-TUNMAMG-*`). Traceable objects make audits, troubleshooting, and the included scripts reliable.

• **Keep assignment groups identical** across the three policies for a platform. Mismatched targeting is the most common reason "the tunnel doesn't auto-connect."

Government correctness [GOV]

• **Always install the gateway with** `intune_env=FXP`**.** Bake it into your build automation so no one can forget. A gateway registered without it lands in Commercial and silently never appears in `intune.microsoft.us`.

• **Standardize on** `*.us` **everywhere** — portals (`intune.microsoft.us`, `portal.azure.us`), sign-in (`login.microsoftonline.us`), and Graph (`graph.microsoft.us` / `dod-graph.microsoft.us`). Update bookmarks, runbooks, and any automation copied from a Commercial project.

• **Pin the right Graph environment in scripts:** `-Environment USGov` for GCC High, `-Environment USGovDoD` for DoD. The scripts in this package default to USGov — change the parameter for DoD.

• **Firewall the Government endpoints.** Allow the **US Government** Office 365/Intune URL & IP ranges, not the Commercial set. Re-verify the published Government endpoint list periodically.

• **Federated tenants:** most Government tenants federate. Configure the Edge/LOB `bypassedUrls` value with your **federation STS host** or first-time sign-in through the tunnel will fail.

Security & compliance [GOV]

• **Treat no-FIPS as a design constraint, not a footnote.** Tunnel for MAM does not support FIPS on Android or iOS. Get written ISSO/ISSM acceptance for the data classification before go-live; if FIPS is mandated for this path, choose a different control.

• **STIG-harden the Linux host** to the DISA MDM/EMM baseline and enable **auditd** (install `mst.rules`) so tunnel activity is logged for your SIEM.

• **Least privilege:** operators hold **Intune Administrator**, not Global Administrator. The gateway registers with an interactive sign-in once; thereafter it uses app credentials — no stored user passwords.

• **Conditional Access:** gate tunnel access with CA (device/user risk, compliant app, MFA). Note the known iOS issue: with an MFA grant on the Tunnel Gateway CA policy, LOB apps must implement `onTokenRequiredWithCallback` — coordinate with developers.

• **Rootless Podman** on RHEL 8.8+ reduces container-escape blast radius (throughput ceiling ~230 Mbps per rootless server — size accordingly).

Certificates & PKI [GOV]

• **Do not assume Cloud PKI.** It is **GCC High only and "planned"** (not GA in DoD; verify in GCC High). Issue the TLS server cert from an **enterprise or supported public CA**.

• **SAN must match** the public FQDN/IP of the Site exactly. Mind lifetimes: iOS public certs ≤ **398 days**.

• **Deploy full chains.** For private-CA-protected intranet resources, put **root + all intermediates** in `CERT-TUNMAMG-Root`; missing intermediates cause Android/iOS server-trust failures.

• **Automate rotation:** replace `site.crt`/`site.key` (or `site.pfx`) → `mst-cli import_cert` → `mst-cli server restart`. Use `import_cert delay <minutes>` to stage a cutover.

Networking & performance

• **Split-tunnel deliberately.** Include only the intranet ranges BYOD users need; keep general internet off the tunnel to protect capacity and privacy. Never use a `0.0.0.0` include/exclude.

• **Add corporate DNS** to the Server configuration **and** to split-tunnel include rules, or internal name resolution fails.

• **Scale horizontally.** Add servers to the same Site behind a load balancer; ~4 CPU / 4 GB / 30 GB per ~1,000 devices is Microsoft's starting point — measure and grow.

• **UDP on** (DTLS) for best performance; only disable UDP (force TCP) if your network blocks it, and only after clients use Defender as the tunnel client.

Platform gotchas

• **One tunnel model per device.** MDM Tunnel and MAM Tunnel must not coexist on the same device — segregate groups.

• **Android:** no Always-on VPN with MAM Tunnel; use Strict Tunnel Mode instead. One Defender app-config policy, one Site. The required apps often won't appear in Company Portal on unenrolled devices — tell users to install manually.

• **iOS:** Per-App only, SDK-delivered, invisible in iOS Settings. Edge on iOS does not support Kerberos or cert-based auth to on-prem sites; native WebSocket APIs are unsupported (JavaScript WebSockets in web views work).

Operations

• **Monitor Health status** and internal-network-accessibility per site; alert on server offline.

• **Stay within one version** of the current Tunnel release; use maintenance windows to control upgrades.

- **Re-run the scripts** on a cadence: `TUNMAMG-Validation.ps1` after any change, `TUNMAMG-Report.ps1` for periodic evidence, `TUNMAMG-Discovery.ps1` when onboarding a new admin.
- **Document your deltas.** Record any place you had to "verify for GCC High/DoD" and what you found — Microsoft updates Government parity often; your notes are the next admin's shortcut.

Verify all GCC-High/DoD specifics against current Microsoft US Government documentation at deployment time.

Deliverables Summary — Microsoft Tunnel for MAM — Government (GCC High) Edition

Solution code: TUNMAMG · **Platform:** Microsoft Intune · **Cloud:** US Sovereign (GCC High / DoD)

Everything included in this solution package and what each piece is for.

Documents (source)

File	Purpose
`executive-summary.md`	Business framing, the availability finding, and the no-FIPS caveat for decision-makers.
`01-solution-overview.md`	What Tunnel for MAM is on GCC High, the **Commercial vs GCC High — what's different** comparison, and the availability caveat up front.
`02-requirements.md`	Every prerequisite (R1–R11) with **verify** + **implement** steps and the object naming convention.
`03-implementation.md`	Ordered, click-by-click steps in the `*.us` portals — gateway install with `intune_env=FXP`, server config, site, and the three MAM policies per platform — with Government caveats inline.
`05-testing.md`	Automated (control-plane) + manual (data-plane) tests with explicit **pass/fail gates**, plus Government-specific verification.
`06-troubleshooting.md`	Symptom → cause → fix, weighted toward GCC-High-specific gotchas.
`07-best-practices.md`	Operational guidance; Government-specific items marked **[GOV]**.
`gcc-high-parity-notes.md`	Cited Commercial-vs-Government parity reference (availability, FIPS, portals, Graph endpoints, Cloud PKI, licensing).
`08-deliverables-summary.md`	This file.

(Category-level prerequisites are published once in `global-requirements-intune-solutions.md`.)

PowerShell scripts (`scripts/`)

All connect to **US Government Microsoft Graph** (`-Environment USGov`; pass DoD parameters for `USGovDoD`). Discovery/Validation/Report are **read-only**; Remediation and Set-ScopeTargeting default to `-WhatIf` (no change until you add `-Commit`).

Script	Type	What it does
`TUNMAMG-Discovery.ps1`	Read-only	Inventories the Government tenant: Tunnel sites/server configs, Tunnel-for-MAM app config/protection policies, target groups, and confirms the Graph context is a US Gov endpoint.
`TUNMAMG-Validation.ps1`	Read-only	Checks the deployment against this solution's expected state (gates A1–A9 in `05-testing.md`): site/server present & online, all three policies per platform exist and share assignment groups, single Defender policy, no MDM/MAM collision, licensing present.
`TUNMAMG-Report.ps1`	Read-only	Produces a point-in-time summary/evidence report (console + optional CSV/JSON) of the Tunnel-for-MAM configuration for audits and change records.
`TUNMAMG-Remediation.ps1`	Safe write (`-WhatIf` default)	Proposes and (with `-Commit`) applies safe fixes — e.g., aligning assignment groups across the three policies, flagging duplicate Defender policies. Never deletes gateway infrastructure.
`TUNMAMG-Set-ScopeTargeting.ps1`	Safe write (`-WhatIf` default)	Creates/populates the pilot and production groups (`GRP-TUNMAMG-Pilot` / `GRP-TUNMAMG-Prod`) and can align policy assignments to them.

Run pattern:

```
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Discovery.ps1
```

```
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Validation.ps1
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Report.ps1
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Set-ScopeTargeting.ps1   # add -Commit to apply
powershell -ExecutionPolicy Bypass -File .\scripts\TUNMAMG-Remediation.ps1          # add -Commit to apply
```

Named objects this solution creates

`SC-TUNMAMG-Gov` (server config) · `SITE-TUNMAMG-Gov` (site) · `AC-TUNMAMG-Defender-Android`, `AC-TUNMAMG-Edge-Android`, `AP-TUNMAMG-Edge-Android` (Android policies) · `AC-TUNMAMG-Edge-iOS`, `AP-TUNMAMG-iOS` (iOS policies) · `CERT-TUNMAMG-Root` (trusted root, if needed) · `GRP-TUNMAMG-Pilot`, `GRP-TUNMAMG-Prod` (groups).

Deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand**, with a go/no-go gate between phases (including a Government compliance gate for the no-FIPS decision).

White-label solution. Verify all GCC-High/DoD specifics against current Microsoft US Government documentation at deployment time.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

TUNMAMG-Discovery.ps1

Purpose: Read-only discovery for Microsoft Tunnel for MAM on a US sovereign-cloud (GCC High / DoD) Intune tenant. Inventories Tunnel sites, server configurations, Tunnel-for-MAM app configuration and app protection policies, and target groups. Confirms the Microsoft Graph context is a US Government endpoint. Replace placeholders like your-tenant-id with real values. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Discovery.ps1 (Graph scopes: DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All, Group.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS A console inventory of Tunnel sites/server configs and MAM policies. The admin sees whether the expected objects exist. PASS = Graph context host ends in .us and the inventory returns. FAIL = Commercial (.com) context, or a Graph connection error. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Discovery.ps1 (Graph scopes: DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All, Group.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS A console inventory of Tunnel sites/server configs and MAM policies. The admin sees whether the expected objects exist. PASS = Graph context host ends in .us and the inventory returns. FAIL = Commercial (.com) context, or a Graph connection error. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

TUNMAMG-Remediation.ps1

Purpose: Proposes and (only with -Commit) applies safe fixes to a Microsoft Tunnel for MAM deployment on a US sovereign-cloud (GCC High / DoD) tenant. It aligns assignment groups across the three Android MAM policies to a chosen group and flags duplicate Defender app-config policies. It never deletes Tunnel gateway infrastructure, sites, or server configurations. Defaults to -WhatIf: nothing changes until you pass -Commit. Replace placeholders like your-tenant-id and the pilot group name with real values. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Remediation.ps1 # preview (WhatIf) powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Remediation.ps1 -Commit # apply (Graph scopes: DeviceManagementApps.ReadWrite.All, Group.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS A list of proposed actions. Without -Commit, no change is made (WhatIf). With -Commit, the approved actions are applied and confirmed. PASS = actions previewed/applied cleanly. .NOTES White-label — the repeatable way. Read-only unless -Commit is supplied; targets Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Remediation.ps1 # preview (WhatIf) powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Remediation.ps1 -Commit # apply (Graph scopes: DeviceManagementApps.ReadWrite.All, Group.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS A list of proposed actions. Without -Commit, no change is made (WhatIf). With -Commit, the approved actions are applied and confirmed. PASS = actions previewed/applied cleanly. .NOTES White-label — the repeatable way. Read-only unless -Commit is supplied; targets Microsoft Graph US Gov endpoints. #>

TUNMAMG-Report.ps1

Purpose: Read-only reporting for Microsoft Tunnel for MAM on a US sovereign-cloud (GCC High / DoD) tenant. Summarizes Tunnel sites, server configurations, server health, the Tunnel-for-MAM app configuration and app protection policies, and target groups into a console summary and optional CSV/JSON export for audit evidence and change records. Replace placeholders like your-tenant-id with real values. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Report.ps1 powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Report.ps1 -OutputPath .\TUNMAMG-report (Graph scopes: DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All, Group.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS A formatted console report. With -OutputPath, a .csv and .json are written next to that path. PASS = report generated. FAIL = Graph connection error. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Report.ps1 powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Report.ps1 -OutputPath .\TUNMAMG-report (Graph scopes: DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All, Group.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS A formatted console report. With -OutputPath, a .csv and .json are written next to that path. PASS = report generated. FAIL = Graph connection error. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

TUNMAMG-Set-ScopeTargeting.ps1

Purpose: Builds the scope-targeting groups this solution uses (GRP-TUNMAMG-Pilot and GRP-TUNMAMG-Prod) on a US sovereign-cloud (GCC High / DoD) tenant, and can add members from a supplied list of user principal names. Use it to stand up the pilot group before Phase 3 and the production group before Phase 5. Defaults to -WhatIf: nothing changes until you pass -Commit. Replace placeholders like your-tenant-id and the member UPNs with real values. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Set-ScopeTargeting.ps1 -Scope Pilot # preview powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Set-ScopeTargeting.ps1 -Scope Pilot -Members user1@your-domain.us,user2@your-domain.us -Commit (Graph scopes: Group.ReadWrite.All, User.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS Reports whether the group exists (creates it if -Commit) and which members would be/were added. PASS = group present and members reconciled. FAIL = Graph error or non-Gov context. .NOTES White-label — the repeatable way. Read-only unless -Commit is supplied; targets Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Set-ScopeTargeting.ps1 -Scope Pilot # preview powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Set-ScopeTargeting.ps1 -Scope Pilot -Members user1@your-domain.us,user2@your-domain.us -Commit (Graph scopes: Group.ReadWrite.All, User.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS Reports whether the group exists (creates it if -Commit) and which members would be/were added. PASS = group present and members reconciled. FAIL = Graph error or non-Gov context. .NOTES White-label — the repeatable way. Read-only unless -Commit is supplied; targets Microsoft Graph US Gov endpoints. #>

TUNMAMG-Validation.ps1

Purpose: Read-only validation of Microsoft Tunnel for MAM against this solution's expected state on a US sovereign-cloud (GCC High / DoD) tenant. Checks that the Tunnel site and server config exist, a server is

registered, the three MAM policies per platform exist and share the same assignment groups, only one Defender app-config policy is present, and there is no MDM/MAM collision. Replace placeholders like your-tenant-id with real values. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Validation.ps1 (Graph scopes: DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All, Group.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS A pass/fail table for gates A1-A9. PASS = all expected objects present and consistent. FAIL = a listed check failed, with a one-line reason and the doc reference to fix it. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\TUNMAMG-Validation.ps1 (Graph scopes: DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All, Group.Read.All. Uses -Environment USGov for GCC High; pass -Environment USGovDoD for DoD.) EXPECTED RESULTS A pass/fail table for gates A1-A9. PASS = all expected objects present and consistent. FAIL = a listed check failed, with a one-line reason and the doc reference to fix it. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>